Dry Firms, Deep Recessions: Corporate Payouts and Aggregate Dynamics*

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Abstract

This paper studies the macroeconomic implications of corporate liquidity holdings over the business cycle. Developing a heterogeneous-firm business-cycle model with precautionary savings, we show that firm-level nonlinear cash holdings drive state dependence in aggregate real dynamics. Specifically, low corporate liquidity triggers sharp net-payout cuts, nearly tripling the consumption drop in response to adverse productivity shocks relative to periods of typical (median) liquidity. We summarize this transmission with the marginal propensity to pay out (MPPO), the share of an additional dollar of firm liquidity passed through to households. The same statistic governs fiscal transmission: tax-financed transfers raise output most when firms are liquid but still accumulating cash, because firms retain the transfer instead of immediately paying it out. Using policy-induced liquidity shocks from net operating loss carryback reforms, we estimate the MPPO across firms and show that the model closely matches its empirical profile.

Keywords: Business cycle, corporate cash holdings, corporate payouts, state dependence, fiscal policy.

JEL codes: E32, E44, G35.

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1 Introduction

U.S. corporate liquidity has increased markedly over recent decades. Measured relative to nominal GDP, aggregate corporate liquid assets more than doubled from around 5% in the 1970s to around 13% in the 2020s.[1] While the drivers of this accumulation of liquid assets are increasingly documented, a fundamental macroeconomic question remains: How does corporate liquidity shape aggregate fluctuations? Does it act as a stabilizing buffer against shocks, or can the state of corporate liquidity itself become a source of macroeconomic amplification? This paper fills that gap. We develop a heterogeneous-firm business-cycle model in which firms accumulate liquidity for precautionary reasons and quantify how firms' liquidity positions shape the transmission of aggregate shocks.

Using this framework, we make three contributions to the business cycle literature. First, we show that corporate financial frictions can generate endogenous state dependence in aggregate dynamics. In contrast to a wide range of heterogeneous-agent models where individual nonlinearities have limited aggregate effects, the nonlinearity in firms' liquidity policies survives aggregation in general equilibrium in our framework. As a result, the response to aggregate productivity shocks is asymmetric and depends on the liquidity distribution of the economy. When firms face an adverse productivity shock with little liquidity, their financing constraints leave them unable to absorb the shock by drawing down cash, forcing them instead to reduce net payouts sharply. By transmitting more of the shock to households through lower payouts, this response amplifies the decline in consumption, which is nearly three times as large as in periods of typical (median) liquidity. Conversely, ample aggregate liquidity dampens the transmission of shocks by allowing firms to smooth net payouts.[2]

Second, building on this state-dependent payout mechanism, we introduce the *marginal propensity to pay out* (MPPO) — the share of an additional unit of liquidity that is immediately distributed to shareholders — as a key determinant of fiscal policy effectiveness in downturns. Because the MPPO determines whether transfers remain on corporate balance sheets or flow back to households, their aggregate effects depend critically on the distribution of firms across liquidity-policy regimes. In particular, the output multiplier of tax-financed transfers to firms is largest in relatively liquid states, where more firms operate in the active cash-accumulation region of their liquidity policy. In this region,

[1] We use the Federal Reserve Board's Financial Accounts core measure of liquid assets. Appendix A.1 defines the measure and also compares it with a broader alternative.

[2] Throughout the model, d denotes net payout, defined empirically as dividends plus share repurchases less equity issuance. For brevity, we occasionally refer to d as dividends. When $d < 0$, however, this denotes net equity issuance rather than a negative cash dividend. The exception is the household evidence in Section 5.2, which records dividend income only.

firms have a low MPPO and retain a substantial share of the transfer rather than immediately passing it through to households. This temporary shift of resources onto corporate balance sheets generates a negative household wealth effect, increasing labor supply and output. By contrast, in low-liquidity states, many firms are at the cash constraint and have a high MPPO, so transfers quickly return to households through payouts and have little effect on labor supply or output.

Third, we provide estimates of the MPPO using firm-level Compustat data and validate the model's transmission mechanism with micro data. Exploiting policy-induced variation in liquidity from the 2002 and 2009 expansions of net operating loss carrybacks, we estimate the change in firms' net payouts generated by an additional dollar of liquidity across the profitability distribution. The estimated MPPO declines sharply with profitability: the difference between the lowest- and highest-profitability quartiles is 0.42, compared with 0.40 in the calibrated model. Without targeting these moments, the model also reproduces three empirical patterns: cash-rich firms smooth dividends more during downturns, liquidity declines with firm scale, and liquidity dispersion narrows with scale. To establish the household-side link in our transmission mechanism, we use PSID data to test whether dividend income reaches household spending. Conditional on labor income, wealth, and household and year fixed effects, a one-dollar increase in dividend income is associated with 7.85-cents of higher subsequent expenditure.

All three results can be traced to a non-monotonicity in firms' cross-sectional liquidity policies. The core friction in our framework is a convex cost of external equity issuance, which creates a precautionary motive for firms to hold liquidity. As in Hennessy and Whited (2007) and Alfaro et al. (2024), the firms' precautionary motive leads to nonlinear saving patterns: liquidity reduces expected financing costs when cash is scarce, but its marginal benefit declines as buffers build, so firms stop accumulating at a finite target. We show that two competing forces govern where firms sit relative to that target. First, the *precautionary target* decreases in firm productivity: persistent high revenues reduce expected financing costs, lowering the desired buffer. Second, *internal cash flow* increases in productivity, expanding the firm's capacity to accumulate. For firms that start with little cash, the interaction of these forces generates a hump-shaped relationship between idiosyncratic productivity and next-period liquidity.

This interaction partitions firms into three endogenous behavioral regimes. Low-productivity firms face high precautionary demand but have insufficient internal cash flow to build their desired buffers. Their limited accumulation is an outcome of the firm's optimization problem: accumulating additional liquidity would require sacrificing scarce current resources or raising costly external equity. Intermediate-productivity firms have sufficient cash flow to actively accumulate, while high-productivity firms have reached a low target and distribute all marginal resources. The firm-level MPPO is therefore high near the cash constraint, low in the active accumulation region, and high again at the target.[3] Aggregate shocks move firms across these regimes, so the composition of the firm distribution — summarized by the average MPPO — shifts endogenously over the business cycle.

Crucially, this composition shapes aggregate outcomes through firms' payout decisions. Although households own firms and trade bonds and equity, these portfolio choices cannot undo changes in aggregate payout timing: bonds are in zero net supply, and equity revaluations do not create current resources for the household sector as a whole. When firms accumulate liquidity, current goods are carried forward and become available only later, reducing current household consumption and, through the wealth effect, affecting labor supply. A distinctive result of our model is that this timing channel remains quantitatively important in general equilibrium. Because firms cannot pool internal liquidity and liquid assets are supplied elastically at a frictional return, there is limited price adjustment to smooth firms' nonlinear policies. This contrasts with Khan and Thomas (2008), where equilibrium movements in wages and interest rates largely offset plant-level nonlinearities, and allows firm-level liquidity decisions to generate aggregate state dependence.

Related Literature. Our paper relates to three branches of the literature. First, we contribute to work on how financial frictions shape aggregate fluctuations. Classic models emphasize amplification through borrowers' net worth and collateral values or through the balance sheets of financial intermediaries (Bernanke et al., 1999; Kiyotaki and Moore, 1997; Gertler and Kiyotaki, 2010). More recent work studies which firms drive this transmission. Investment responses to monetary policy vary with default risk and firms' age and dividend status (Ottonello and Winberry, 2020; Cloyne et al., 2023); financing choices vary with firm size (Begenau and Salomao, 2018); and constraints across the firm age and size distributions have important aggregate implications (Kochen, 2022; Ferreira et al., 2023). Credit shocks can also generate large fluctuations by disrupting the allocation of capital across heterogeneous producers (Khan and Thomas, 2013). Much of this literature emphasizes investment, capital allocation, and firm dynamics. We identify a complementary payout channel through which financial frictions shape aggregate fluctuations: by inducing precautionary liquidity accumulation, these frictions make firms' payouts

[3] In the calibrated ergodic distribution, only the top 2% of firms lie on the upward-sloping branch of the MPPO schedule, so the quartile-level MPPO declines with profitability, as in our empirical estimates.

state-dependent and thus aggregate quantities.

Second, we contribute to the literature on corporate liquidity. Opler et al. (1999) document that firms with riskier cash flows and less access to external finance hold more cash. Acharya et al. (2007) analyze cash and debt as alternative hedging instruments, showing that constrained firms save cash when income shortfalls coincide with investment opportunities. Almeida et al. (2004) find that constrained firms exhibit a positive cash-flow sensitivity of cash, although Riddick and Whited (2009) caution that this sensitivity also reflects income uncertainty, investment responses, taxation, and external-finance costs. Regarding long-run trends, Bates et al. (2009) identify greater cash-flow risk as a contributor to rising U.S. cash ratios, Chen et al. (2017) link the distinct global rise in corporate saving to profits outpacing dividends, and Armenter and Hnatkovska (2017) attribute a substantial share of firms' savings to the tax treatment of debt and equity. At cyclical frequencies, Alfaro et al. (2024) show that uncertainty shocks raise cash holdings while reducing investment and payouts. Related work links liquidity to employment. Bacchetta et al. (2019) show that adverse external-liquidity shocks raise cash ratios and reduce employment, while Melcangi (2024) shows how precautionary cash accumulation shapes employment during credit tightening. We complement this literature by showing how financing frictions generate precautionary liquidity that smooths payouts, producing state-dependent consumption responses to real shocks.

Third, we relate to heterogeneous-agent business-cycle models and state-dependent dynamics in the aggregate. Aiyagari (1994) provides the canonical incomplete-markets model of precautionary household saving, while Krusell and Smith (1998) show that its aggregate dynamics can be summarized remarkably well by mean capital. On the firm side, Khan and Thomas (2008) show that nonconvex adjustment costs generate nonlinear plant-level investment policies, but equilibrium movements in prices eliminate aggregate nonlinearities. We show that nonlinear corporate liquidity policies can instead survive aggregation and generate state dependence in aggregate dynamics.

Structure The rest of the paper is organized as follows. Section 2 illustrates the mechanism, Section 3 presents the quantitative model, and Section 4 characterizes firms' nonlinear liquidity policies and the MPPO. Section 5 validates the transmission mechanism with firm- and household-level evidence and quantifies its aggregate implications, including fiscal transmission. Section 6 concludes.

2 A simple theory of cash holdings and payouts

We begin with a simple two-period model that analytically illustrates the main mechanism of the paper. Costly external equity issuance creates a precautionary motive for firms to hold liquidity, while its low return limits accumulation.[4] The resulting cash policy is nonlinear, and payout behaviour inherits this nonlinearity, so the transmission of productivity shocks to household consumption depends on where firms sit in the liquidity distribution.

Household The household lives for two periods, supplies labor inelastically, owns a continuum of identical firms, saves in a risk-free bond b at price $q = 1/(1+r)$ that is in zero net supply, and receives labor income w_t, the firm's payout net of issuance costs, $d_t - \mathcal{C}(d_t)$, and a lump-sum rebate R_t in each period. Period utility u is strictly increasing and strictly concave, and consumption c_t is strictly positive in both periods. The household's problem is:

$$\begin{aligned} \max_{c_1, c_2(\cdot), b} \quad & u(c_1) + \beta\, \mathbb{E}\left[u\left(c_2(A_2)\right)\right], \\ \text{subject to:} \quad & c_1 + qb = w_1 + [d_1 - \mathcal{C}(d_1)] + R_1, \\ & c_2(A_2) = w_2 + [d_2(A_2) - \mathcal{C}\left(d_2(A_2)\right)] + R_2(A_2) + b, \end{aligned}$$

where β is the subjective discount factor and A_2 denotes stochastic aggregate productivity in period 2. Variables that depend on the period-2 realization are written as explicit functions of A_2, which is unknown when period-1 decisions are taken. In equilibrium, the rebate satisfies $R_t = \mathcal{C}(d_t)$. The bond Euler equation gives $q = \mathbb{E}[\mathcal{M}(A_2)]$, where $\mathcal{M}(A_2) = \beta\, u'(c_2(A_2))\,/u'(c_1)$ is the stochastic discount factor. Further, bond market clearing implies $b = 0$. Together with the rebate condition $R_t = \mathcal{C}(d_t)$, the household's budget constraints reduce to

$$c_1 = w_1 + d_1, \qquad c_2(A_2) = w_2 + d_2(A_2).$$

Because wages are fixed in this simple model, productivity shocks affect household consumption only through the firm's payout policy.

[4]In what follows we will, with a slight abuse of terminology, refer to liquid assets and cash holdings interchangeably.

Representative firm The firm enters period 1 with cash holdings n_1. In each period $t \in \{1, 2\}$ it produces A_t units of output and pays the wage bill w_t. Productivity is drawn independently across periods from

$$A_t = \begin{cases} \bar{A} + \Delta_A & \text{with probability } 1/2, \\ \bar{A} - \Delta_A & \text{with probability } 1/2, \end{cases}$$

so that $A_t \in \{\bar{A} - \Delta_A, \bar{A} + \Delta_A\}$ with $\Delta_A \in (0, \bar{A})$. Period-1 productivity A_1 is realized before the firm chooses; only A_2 is unknown at that point. At the end of period 2 the firm distributes its entire net worth and exits.

Two frictions shape its problem. Negative net payouts require an equity issue at the quadratic cost $\mathcal{C}(d) = \frac{\mu}{2} d^2 \mathbb{I}(d < 0)$.[5] And cash bears a lower return than the risk-free rate: it earns a fraction $\lambda \in (0, 1)$ of the risk-free rate, $r^n = \lambda r$, so its price exceeds the bond price,

$$q^n = \frac{1}{1 + \lambda r} > q,$$

with the same wedge parameter λ as in the quantitative model (Cooley and Quadrini, 2001; Alfaro et al., 2024).[6] We also restrict the magnitude of productivity fluctuations Δ_A as follows:

Assumption 1. *The aggregate productivity spread satisfies* $\Delta_A > |w_2 - \bar{A}|$, *or equivalently,* $\bar{A} - \Delta_A < w_2 < \bar{A} + \Delta_A$.

Under Assumption 1, a firm that enters period 2 without cash must issue equity when productivity is low, but not when productivity is high: cash matters in one state only. For cash holdings below the issuance-free level, the external-financing cost is therefore active only in the low state. For compactness, we write $\mathcal{M}^L \equiv \mathcal{M}(\bar{A} - \Delta_A)$ for the stochastic discount factor in the low state.

Taking prices and the stochastic discount factor as given, the firm chooses cash $n_2 \geq 0$

[5] We consider a purely quadratic cost, as in the quantitative model of Section 3, so that $\mathcal{C}$ is continuously differentiable on the whole domain, with $\mathcal{C}'(d) = \mu d \, \mathbb{I}(d < 0)$. Appendix B adds a linear component, and Proposition 1 remains valid under that specification.

[6] Cash is a storage technology: the resources $q^n n_2 - n_1$ that the firm devotes to liquidity in period 1 return as n_2 in period 2, at a rate below the risk-free rate. Appendix F microfounds this wedge.

to maximize its value to the shareholder:

$$\begin{aligned}
\max_{n_2 \geq 0} \quad & d_1 - \mathcal{C}(d_1) + \mathbb{E}\left[\mathcal{M}(A_2)\left(d_2(A_2) - \mathcal{C}\left(d_2(A_2)\right)\right)\right], \\
\text{s.t.} \quad & d_1 = A_1 - w_1 - q^n n_2 + n_1, \\
& d_2(A_2) = A_2 - w_2 + n_2, \\
& \mathcal{C}(d) = \frac{\mu}{2} d^2 \mathbb{I}(d < 0).
\end{aligned}$$

Net payouts are residual in both periods, so n_2 is the only choice variable; since $\mathcal{C}$ is convex and the constraints are linear in n_2, the objective is concave and the first-order condition is necessary and sufficient.[7]

We model $\mathcal{C}(d_t)$ as a wedge in the firm's payout problem and rebate the proceeds lump-sum to the household. The rebate does not enter the firm's objective. Thus, the firm optimizes against $d_t - \mathcal{C}(d_t)$, while the household receives $d_t - \mathcal{C}(d_t) + R_t = d_t$ in equilibrium. The wedge therefore distorts the firm's cash and payout choices without absorbing aggregate resources. We impose this convention deliberately to isolate the mechanism of interest.[8]

Costly issuance gives the firm a reason to self-insure: cash carried into period 2 shrinks the equity issue needed when revenue falls short, and with it the expected issuance cost. The next result links the size of the friction to the size of the buffer.

Proposition 1 (Optimal cash holding)**.**
Suppose the firm pays a non-negative net payout in period 1, so that the marginal unit of cash is financed internally. Its optimal cash holding is

$$n_2^* = \max\left\{0,\ \underbrace{w_2 + \Delta_A - \bar{A}}_{\equiv\, n_2^{\max}} - \frac{2\left(q^n - q\right)}{\mu\, \mathcal{M}^L}\right\},$$

which is (weakly) increasing in the equity-issuance-cost parameter μ and bounded above by the issuance-free level $n_2^{\max}$, a bound it approaches as $\mu \to \infty$ or as the return wedge vanishes ($\lambda \to 1$).

Proof. For $n_2 \geq n_2^{\max}$ neither state triggers issuance, so the marginal value of cash to the

[7] As is standard in this class of models, the firm takes $\mathcal{M}$ as given even though $\mathcal{M}$ depends on equilibrium consumption. The comparative statics in Propositions 1 and 2 therefore hold the stochastic discount factor fixed; the quantitative model of Section 3 allows for the general-equilibrium feedback.

[8] The assumption can also be rationalized as share dilution, where $\mathcal{C}(d)$ is the discount at which new equity is sold and the household holds the newly issued claim alongside the existing one, or as fee income accruing to an underwriting sector that the household owns.

firm is $-q^n + q < 0$ and the optimum lies in $[0, n_2^{\max})$. In this region, Assumption 1 implies the firm issues equity in period 2 if and only if productivity is low: $d_2(\bar{A} - \Delta_A) < 0$ and $d_2(\bar{A} + \Delta_A) > 0$. With $d_1 \geq 0$ we have $\mathcal{C}'(d_1) = 0$, so the first-order condition for n_2 is

$$-q^n + q - \tfrac{1}{2}\,\mathcal{M}^L\,\mu\, d_2(\bar{A} - \Delta_A) = 0,$$

which, substituting $d_2(\bar{A} - \Delta_A) = \bar{A} - \Delta_A - w_2 + n_2$, yields the expression above. Because $q^n > q$, the correction term is strictly negative and $n_2^* < n_2^{\max}$. Whenever the solution is interior,

$$\frac{\partial n_2^*}{\partial \mu} = \frac{2\,(q^n - q)}{\mu^2\,\mathcal{M}^L} > 0,$$

and the correction term vanishes as $\mu \to \infty$ or as $q^n \downarrow q$, which by $q^n = 1/(1 + \lambda r)$ corresponds to $\lambda \to 1$. Finally, $n_2^* > 0$ if and only if $\mu\,\mathcal{M}^L\, n_2^{\max} > 2(q^n - q)$: the firm holds cash only when the issuance friction is large relative to the carry cost of liquidity. ■

A higher μ makes external equity more expensive, so the firm carries a larger buffer. At $n_2^{\max}$ the firm can absorb the worst realization without going to the equity market, so an extra unit of cash no longer relaxes the period-2 financing constraint and only costs the carry. The firm nevertheless stops strictly short of that point precisely because cash is return-dominated: its target n_2^* is where the marginal benefit of liquidity meets the carry cost, not where that benefit disappears. Throughout the paper we call a firm sitting at its target *satiated*, and reserve $n_2^{\max}$ for the strictly higher level at which issuance is ruled out altogether. The buffer also rises with $\mathcal{M}^L$: the more the household values resources in the bad state, the more aggressively the firm insures it.[9]

Proposition 1 conditions on a non-negative period-1 dividend. Dropping that restriction characterizes the cash policy over the whole state space, and with it the pass-through of liquidity to consumption.

Proposition 2 (The state-dependent consumption response)**.**
The pass-through of an incremental increase in initial cash into consumption, conditional on a negative TFP realization, lies in the unit interval:

$$1 \geq \frac{c_1(n_1 + \Delta_n, \bar{A} - \Delta_A) - c_1(n_1, \bar{A} - \Delta_A)}{\Delta_n} > 0.$$

[9]This is a statement about $\mathcal{M}^L$, which the firm takes as given. It does not translate directly into a comparative static in risk aversion, since greater curvature of u also moves q and $c_2(\bar{A} - \Delta_A)$ in equilibrium.

Moreover, this pass-through is state-dependent and piecewise determined by the firm's optimal cash policy.

Proof. As consumption equals wages plus dividends, it suffices to characterize the marginal response of net payouts to initial cash holdings. For Δ_n small enough that the firm remains in the same regime, the difference quotient equals $\partial d_1/\partial n_1$. There are three distinct situations:

1. The firm has a negative net payout and raises equity in period 1 and holds a strictly positive cash position:

$$\frac{c_1(n_1+\Delta_n, \bar{A}-\Delta_A) - c_1(n_1, \bar{A}-\Delta_A)}{\Delta_n} = \frac{\partial d_1}{\partial n_1} = 1 - q^n \frac{\partial n_2}{\partial n_1}$$
$$= 1 - \frac{(q^n)^2}{(q^n)^2 + \frac{1}{2}\mathcal{M}^L} = \frac{\frac{1}{2}\mathcal{M}^L}{(q^n)^2 + \frac{1}{2}\mathcal{M}^L} \in (0,1),$$

   where $\partial n_2/\partial n_1 = q^n / \left((q^n)^2 + \frac{1}{2}\mathcal{M}^L\right)$ follows from implicitly differentiating the period-1 first-order condition. This expression is strictly between 0 and 1 since $(q^n)^2 > 0$ and $\mathcal{M}^L > 0$, and it does not depend on μ: the issuance cost determines the level of cash the firm targets, not the fraction of a marginal dollar it retains.[10]

2. If the firm is at the no-borrowing limit, it implements $n_2 = 0$ and

$$\frac{c_1(n_1+\Delta_n, \bar{A}-\Delta_A) - c_1(n_1, \bar{A}-\Delta_A)}{\Delta_n} = 1.$$

3. If the firm is already at the target n_2^* of Proposition 1, then

$$\frac{c_1(n_1+\Delta_n, \bar{A}-\Delta_A) - c_1(n_1, \bar{A}-\Delta_A)}{\Delta_n} = 1.$$

For larger Δ_n the difference quotient is a convex combination of these three values and therefore also lies in the unit interval. ■

This result shows that the pass-through of liquidity into consumption is nonlinear. When firms are either cash-constrained or already at their target, an additional unit of cash is fully paid out, leading to complete pass-through to consumption. In contrast,

[10] If Assumption 1 is relaxed so that the firm issues in both period-2 states, $\frac{1}{2}\mathcal{M}^L$ is replaced by q; the pass-through remains in $(0,1)$ and independent of μ.

when firms operate in the interior of the cash policy function, additional cash is partially retained, dampening the consumption response.

The nonlinearity of the cash policy function therefore generates state-dependent consumption dynamics. A sufficient statistic summarizing this mechanism is the firm's *Marginal Propensity to Pay Out* (MPPO).[11]

Definition 1 (Marginal propensity to pay out)**.**
We define the MPPO as

$$MPPO \equiv \frac{\partial d_t(n_t)}{\partial n_t}.$$

The MPPO measures the increase in net payout generated by an additional unit of predetermined cash; $1 - \text{MPPO}$ is retained. When net payout is negative, a higher payout represents a reduction in equity issuance. Here the relevant object is $\partial d_1 / \partial n_1$ from Proposition 2, and the definition carries over unchanged to the quantitative model.

The MPPO connects the model directly to a large empirical literature studying how firms adjust payouts in response to liquidity shocks. Early evidence by Fazzari et al. (1988) and Almeida et al. (2004) shows that financially constrained firms exhibit a higher propensity to retain internal funds, consistent with precautionary saving motives. More recently, Almeida and Campello (2010) and Acharya et al. (2013) document that firms' cash retention rises in periods of heightened uncertainty and tighter financing conditions, further indicating that payout responses to liquidity shocks are state-dependent. These empirical estimates of payout sensitivities — whether based on cash flow shocks, tax windfalls, or credit supply shifts — can therefore be interpreted as reduced-form estimates of the MPPO.

Corollary 1 (MPPO as the marginal consumption pass-through)**.**
At the margin, the state-dependent consumption pass-through of Proposition 2 equals the firm's Marginal Propensity to Pay Out: taking the limit as $\Delta_n \to 0$,

$$\frac{c_1(n_1 + \Delta_n, \bar{A} - \Delta_A) - c_1(n_1, \bar{A} - \Delta_A)}{\Delta_n} = MPPO.$$

Proof. The proof of Proposition 2 shows that the difference quotient equals $\partial d_1 / \partial n_1$ in each regime. Taking $\Delta_n \to 0$ yields the derivative $\partial c_1 / \partial n_1 = \partial d_1 / \partial n_1$, since $c_1 = w_1 + d_1$ with the wage w_1 predetermined; this is the MPPO of Definition 1. ■

[11] The MPPO is a sufficient statistic for the pass-through of financial conditions to aggregate consumption in this simplified setup. In the quantitative model of Section 3 two additional margins affect the consumption response: labor supply and equity price fluctuations.

A marginal unit of corporate liquidity thus transmits to household consumption entirely through the firm's payout response, so the MPPO summarizes how corporate liquidity passes through to consumption. The reduced-form payout sensitivities discussed above therefore map directly into consumption pass-throughs in the model. The model makes the MPPO endogenous: it varies with firms' cash positions relative to their financing constraints and targets. This nonlinearity governs how payout behavior transmits financial conditions to aggregate consumption.

In contrast, when cash is held by households rather than firms, optimal saving equates the marginal utility of consumption today and tomorrow, subject to the cost of cash holdings. The household saving policy is then smooth except at the borrowing constraint, which delivers approximate aggregation and has been shown to be insufficient to generate aggregate state dependence in consumption (Krusell and Smith, 1998). By shifting liquidity to firms' balance sheets, our framework introduces a nonlinear cash-holding policy even away from borrowing constraints, which translates into state-dependent payout behavior and consumption dynamics.

3 Quantitative model

To assess the quantitative implications of our mechanism over the business cycle, we now embed it in a general-equilibrium model with heterogeneous firms. Firms face idiosyncratic and aggregate productivity risk, choose labor and net payouts, and use cash to insure against costly external equity issuance, in the spirit of Froot et al. (1993). A representative household supplies labor, owns the firms, and trades a risk-free bond in zero net supply. The government can levy lump-sum taxes on the household and transfer the proceeds to firms. The cross-sectional distribution of firms over productivity and cash holdings is an endogenous aggregate state. Its evolution determines how many firms are constrained, accumulating cash, or at their cash targets — and therefore how aggregate payouts and consumption respond to shocks.

3.1 Production and aggregate states

Firms use labor to produce the final good. The production function of firm i has decreasing returns to scale and the following functional form:

$$y_{it} = A_t z_{it} l_{it}^{\gamma},$$

where y_{it} denotes final good output, l_{it} labor demand, $\gamma < 1$ is the span-of-control parameter, and z_{it} and A_t are idiosyncratic and aggregate productivity, respectively.[12] Idiosyncratic productivity follows

$$\log z_{it} = \rho_z \log z_{i,t-1} + \epsilon_{it}, \qquad \epsilon_{it} \overset{i.i.d.}{\sim} N(0, \sigma_z^2),$$

where ρ_z and σ_z^2 are the persistence and innovation variance. For computation, we approximate this process by a finite-state Markov chain on $\mathbb{Z}$. Aggregate productivity follows a two-state Markov process. Its transition matrix and support are

$$\Gamma_A = \begin{bmatrix} p(A_{t+1} = A_B \mid A_t = A_B) & 1 - p(A_{t+1} = A_B \mid A_t = A_B) \\ 1 - p(A_{t+1} = A_G \mid A_t = A_G) & p(A_{t+1} = A_G \mid A_t = A_G) \end{bmatrix},$$
$$A_t \in \{A_B, A_G\}, \qquad A_B := \bar{A} - \Delta_A = 1 - \Delta_A, \quad A_G := \bar{A} + \Delta_A = 1 + \Delta_A,$$

where we normalize mean aggregate productivity to $\bar{A} = 1$ and calibrate Δ_A to match aggregate output volatility.[13]

Due to aggregate risk and incomplete markets, the distribution of firms over idiosyncratic cash holdings n_{it} and productivity z_{it} varies over time. Let $\Phi_t(n_{it}, z_{it})$ denote this distribution. Given firms' optimal cash policy and the transition process for idiosyncratic productivity, it evolves according to

$$\Phi_{t+1}(n_{i,t+1}, z_{i,t+1}) = \mathbf{G}(\Phi_t(n_{it}, z_{it}), A_t; N),$$

where $\mathbf{G}$ is the transition operator induced jointly by firms' optimal cash policy and the exogenous transition process for idiosyncratic productivity. It maps the current firm distribution and aggregate productivity into the next-period distribution. Finally, for ease of notation, denote the collection of aggregate state variables X_t as

$$X_t \equiv \{\Phi_t(n_{it}, z_{it}), A_t\}.$$

[12] We abstract from capital accumulation to isolate the cash–payout mechanism. The decreasing-returns technology can be interpreted as a reduced-form representation of production after optimizing over other variable inputs.

[13] The two-state approximation does not materially affect the results. The repeated-transition method accommodates finer grids; we use two states to distinguish booms from recessions transparently. For example, Lee (2026b) uses five aggregate states.

3.2 Cash holdings and financial frictions

Payouts and cash holdings Firms generate profits equal to revenues net of the wage bill and a fixed operating cost $\xi > 0$ in each period. They then decide how much to distribute as net payouts d_{it} to their ultimate owners, the households. The portion of earnings retained after payouts is allocated to adjusting the firm's cash balance. We define net investment in cash as

$$h_{it} \equiv q_t^n n_{i,t+1} - n_{it},$$

where n_{it} denotes the firm's cash holdings and q_t^n is the price of one unit of cash delivered at $t+1$. Firms face a no-borrowing constraint on cash, $n_{i,t+1} \geq 0$. This mirrors the borrowing limits in standard incomplete-markets models such as Aiyagari (1994) and Huggett (1993).

Further, we assume that the return on cash is lower than the risk-free rate, following Cooley and Quadrini (2001), Jeenas (2023), and Alfaro et al. (2024). Specifically, we parameterize the relative net return on cash by $\lambda \in (0,1)$:

$$r_t^n = \lambda r_t.$$

Equivalently,

$$q_t^n = \kappa_t q_t, \qquad \kappa_t \equiv \frac{1+r_t}{1+\lambda r_t} > 1,$$

where q_t is the risk-free bond price. As in the two-period model, $\lambda < 1$ makes cash less attractive than the risk-free asset as a store of value.

On the aggregate level, the net supply of liquid assets is assumed to follow an exogenous process linked to the price of liquid assets q_t^n. In particular, we again follow Alfaro et al. (2024) and impose a constant elasticity of supply specification:

$$N_{t+1}^S = \Omega \left(\frac{q_t^n}{\kappa_t q_t} \right)^{1/\zeta},$$

where ζ is the inverse elasticity of the liquid asset supply, and $\Omega > 0$ is a scaling constant. For a microfoundation of the market for cash and the wedge in the price of cash, see Jeenas (2023) and Appendix F.

External financing cost As in the two-period model, firms face a non-negativity constraint on cash holdings and incur an external financing cost $\mathcal{C}(d_{it})$ whenever net payouts are negative, in the spirit of Jermann and Quadrini (2012) and Riddick and Whited

(2009).[14] We assume that this cost takes the following functional form:

$$\mathcal{C}(d_{it}) = \frac{\mu}{2}\mathbb{I}\{d_{it} < 0\}d_{it}^2,$$

where μ governs the magnitude of the financing cost, determining how costly it is for firms to raise external funds when net payouts are negative. A higher μ implies that even small equity issuances result in significant costs, discouraging firms from relying on external financing and strengthening their precautionary savings motive. The firm's current payoff net of equity issuance cost is given by

$$g(d) \equiv d_{it} - \frac{\mu}{2}\mathbb{I}\{d_{it} < 0\}d_{it}^2.$$

This function is continuously differentiable ($\mathbb{C}^1$) and concave, ensuring smooth adjustment at $d_{it} = 0$ without kinks.[15,16]

The external financing cost captures firms' limited ability to adjust funding sources in response to financial conditions, as in Jermann and Quadrini (2012). This also aligns with the empirical literature on dividend smoothing where firms avoid drastic payout fluctuations due to managerial incentives and agency considerations (Leary and Michaely, 2011; Bliss et al., 2015). In particular, Leary and Michaely (2011) show that cash-rich firms smooth dividends significantly more than others, a pattern the model replicates. Together, the return wedge and external financing cost generate the same precautionary cash-holding mechanism as in the two-period model: firms use cash to cushion adverse productivity realizations and thereby partially smooth net payouts.

3.3 Recursive firm problem

The firm's problem can now be written in recursive form. Suppressing time and firm subscripts, the firm observes its idiosyncratic states (n, z) and the aggregate state $X = (\Phi, A)$. Labor is chosen statically, while the current net payout d and next-period cash

[14] In line with the two-period model in Section 2, the equity issuance cost is rebated back to the household.

[15] The theoretical results do not rely on the convexity of the adjustment cost. Fixed or linear costs preserve the precautionary mechanism, while the convex specification delivers the analytical characterizations used here and is useful computationally because it preserves weak concavity of the firm's value function over the relevant state space.

[16] Jermann and Quadrini (2012) highlight that the convexity assumption aligns with empirical findings by Altinkilic and Hansen (2000) and Hansen and Torregrosa (1992), who document that underwriting fees increase at a rising marginal cost as the offering size grows.

holdings n' are chosen dynamically. Its operating profit is

$$\pi(z;X) \equiv \max_{l} \{zAl^{\gamma} - w(X)l - \xi\}.$$

The recursive problem is then

$$\begin{aligned} J(n,z;X) = \max_{n',d} \quad & g(d) + \mathbb{E}\big[\mathcal{M}(X;X')J(n',z';X') \mid z, A\big] \\ \text{subject to} \quad & d = \pi(z;X) + n - q^{n}(X)n' + T^{f}(X), \\ & n' \geq 0, \\ & \mathcal{C}(d) = \frac{\mu}{2}\mathbb{I}\{d<0\}d^{2}, \\ & X' = (\Phi', A'), \qquad \Phi' = \mathbf{G}^{f}(\Phi, A), \end{aligned} \tag{1}$$

where $w(X)$ is the equilibrium wage, J denotes the value function of the firm, $\mathcal{M}(X;X')$ is the household stochastic discount factor, $g(d)$ is the firm's current payoff net of the equity issuance cost, and $T^{f}(X)$ is a lump-sum transfer to each firm. The optimal next-period cash, net-payout, and labor policies are denoted by $N(n,z;X)$, $D(n,z;X)$, and $L(n,z;X)$, respectively.

3.4 Household

The representative household chooses consumption c, labor supply l^{H}, risk-free bond holdings b', and a next-period equity claim s', chosen contingent on the next aggregate shock, $s' = s'(A')$. It owns the firms, receives the equity-issuance costs as a lump-sum rebate,

$$R(X) \equiv \int \mathcal{C}(D(n,z;X))\, d\Phi, \tag{2}$$

and pays a lump-sum tax $T(X)$. Its problem is

$$\begin{aligned} V^{H}(b,s;X) = \max_{c,s'(\cdot),b',l^{H}} \quad & \log c - \frac{\eta}{1+\frac{1}{\chi}}(l^{H})^{1+\frac{1}{\chi}} + \beta\mathbb{E}\Big[V^{H}(b',s'(A');X') \mid A\Big] \\ \text{subject to} \quad & c + q(X)b' + \sum_{A'} \Gamma_{A,A'}\, \mathcal{M}(X;X')\, s'(A') \\ & \qquad = w(X)l^{H} + s + b + R(X) - T(X), \\ & X' = (\Phi', A'), \qquad \Phi' = \mathbf{G}^{h}(\Phi, A). \end{aligned} \tag{3}$$

Here s denotes the payoff of the household's current equity claim—in equilibrium, the cum-payoff value of the firm sector—and the sum on the left-hand side is the date-t value of the next-period claim. Because issuance costs return to the household through $R(X)$, they distort firms' payout choices without absorbing resources. The parameter η governs labor disutility and χ is the Frisch elasticity. With log utility, household optimality implies

$$\mathcal{M}(X;X') = \beta \frac{C(X)}{C(X')}, \qquad q(X) = \mathbb{E}\left[\mathcal{M}(X;X') \mid A\right].$$

3.5 Government

The government collects the exogenous lump-sum tax $T(X)$ from the household and distributes its proceeds equally among firms. Its budget constraint is

$$T(X) = \int T^f(X)\, d\Phi. \tag{4}$$

Both objects are zero in the baseline calibration; we retain them here for the fiscal-policy exercises in Section 5.4.

3.6 Competitive equilibrium

Definition 2. *A recursive competitive equilibrium is a set of functions*

$$\left(q, q^n, \mathcal{M}, w, T, T^f, R, J, N, H, D, L, V^H, C, L^H, S, B, N^S, \mathbf{G}^f, \mathbf{G}^h, \mathbf{G}, \Phi\right)$$

that solve the firm and household problems, satisfy the government budget constraint, and clear all markets, as described by the following conditions:

1. *Taking q^n, $\mathcal{M}$, w, T^f, and the perceived law of motion $\mathbf{G}^f$ as given, J solves the firm problem* (1)*; (N, D, L) are the associated optimal policy functions, and $H(n,z;X) = q^n(X)N(n,z;X) - n$ is the implied net-cash-investment policy.*

2. *Taking q, w, T, R, and the perceived law of motion $\mathbf{G}^h$ as given, V^H solves the household problem* (3)*, and (C, B, L^H, S) are the associated policy functions.*

3. *Given T, the transfer T^f satisfies the government budget constraint* (4)*, and the rebate R satisfies* (2)*.*

4. *The perceived laws of motion are consistent with the actual law of motion induced by the optimal cash policy N and the exogenous transition process for idiosyncratic productivity:*

$$\mathbf{G}^f(\Phi, A) = \mathbf{G}^h(\Phi, A) = \mathbf{G}(\Phi, A; N).$$

5. *The labor market clears,* $L^H(X) = \int L(n,z;X)\, d\Phi$.

6. *The goods market clears,* $\int y(n,z;X)\, d\Phi = C(X) + \int [\xi + H(n,z;X)]\, d\Phi$.

7. *The market for shares clears state by state: for every realization* A' *of the aggregate shock,*

$$S(X;A') = \int J(n',z';X')\, d\Phi', \qquad X' = \left(\mathbf{G}(\Phi, A; N), A'\right),$$

so the household's claim pays the cum-payoff value of the firm sector in every aggregate state.

8. *The market for next-period liquid assets clears,* $N^S(X)' = \int N(n,z;X)\, d\Phi$.

9. *The bond market clearing condition,* $B(X) = 0$*, is satisfied by Walras' Law.*

4 The role of market incompleteness and financial frictions

We now characterize how market incompleteness and financial frictions shape firms' liquidity policies in the globally solved recursive competitive equilibrium of the quantitative model. The target cash holdings and state-dependent marginal propensity to pay out (MPPO) of Section 2 are now quantitatively analyzed within the equilibrium.[17] Our central result is that the MPPO equals one both at the cash constraint and at the target, but for opposite reasons: a constrained firm passes through marginal resources because cash cannot fall below zero; a firm at its target does so because it optimally chooses not to accumulate more. In between, a firm raising equity retains part of a marginal unit and its MPPO falls below one. Aggregate shocks that shift the distribution across these regions change the economy's average pass-through — the origin of the state-dependent responses we study in Section 5.

4.1 Target cash holdings

As in Section 2, costly external financing makes cash valuable as a precautionary buffer, while its low return makes it costly to hold. This trade-off generates a productivity-specific target beyond which the firm does not accumulate additional cash. We define the firm's cash on hand as[18]

$$m(n,z) \equiv \pi(z) + n.$$

[17] Throughout this section, we suppress the aggregate state X and write $n'(n,z)$, $d(n,z)$, and $MPPO(n,z)$.

[18] Transfers are set to zero in the non-stochastic steady state.

At the non-stochastic steady state, the firm's problem is

$$J(n,z) = \max_{n' \geq 0} \left\{ g\big(m(n,z) - q^n n'\big) + \beta \mathbb{E}[J(n',z') \mid z] \right\}, \tag{5}$$

Proposition 3 (Existence of a target cash-holding level)**.** *Suppose the policy functions are non-trivial: $n'(\tilde{n},z) > 0$ and $d(\tilde{n},z) > 0$ at some $\tilde{n} > 0$, given z. Suppose also that the continuation objective*

$$K_z(n') \equiv -q^n n' + \beta \mathbb{E}[J(n',z') \mid z]$$

attains a unique maximum on $\mathbb{R}_+$.[19] *Then there exists $\overline{n}(z) > 0$ such that*

$$n'(n,z) \leq \overline{n}(z) \qquad \text{for every } n \geq 0.$$

Moreover,

$$n'(n,z) = \overline{n}(z)$$

for every n satisfying $\pi(z) + n \geq q^n \overline{n}(z)$; in particular, for every $n \geq \tilde{n}$. Thus, the cash policy is flat once cash on hand is sufficient to finance the target.

Proof. Using $d = m(n,z) - q^n n'$, the firm's objective can be written as

$$m(n,z) + K_z(n') - \mathcal{C}\big(m(n,z) - q^n n'\big).$$

Let $\overline{n}(z)$ denote the unique maximizer of K_z. For any $n' > \overline{n}(z)$, uniqueness implies $K_z(n') < K_z(\overline{n}(z))$. The implied net payout is also lower, so the equity issuance cost is weakly higher. Hence the target strictly dominates every choice above it, and

$$n'(n,z) \leq \overline{n}(z)$$

for every $n \geq 0$.

Next consider the cash position $\tilde{n}$ in the hypothesis. Because $d(\tilde{n},z) > 0$, the financing cost is zero in a neighborhood of $n'(\tilde{n},z)$. The firm's objective in that neighborhood is therefore $m(\tilde{n},z) + K_z(n')$. Since $n'(\tilde{n},z) > 0$ and is optimal, it is a local maximizer of K_z.

[19] Concavity, finite productivity support and the low return on cash ensure existence. The Bellman problem implies that $J(\cdot,z)$ and hence K_z are continuous and concave. Once cash holdings are high enough to eliminate equity issuance in every continuation state, the envelope condition gives $J_n = 1$, so K_z declines at rate $q^n - \beta > 0$. The maximization can therefore be restricted to a compact interval. Uniqueness is the additional condition: because positive payouts are frictionless, K_z could in principle have a flat maximal segment. The calibrated model has a unique maximizer.

Concavity makes it a global maximizer, and uniqueness implies

$$n'(\widetilde{n}, z) = \overline{n}(z) > 0.$$

Finally, suppose $m(n, z) \geq q^n \overline{n}(z)$. The target then implies a non-negative net payout and no external financing cost. For every $n'' \neq \overline{n}(z)$,

$$m(n, z) + K_z(n'') - \mathcal{C}\left(m(n, z) - q^n n''\right) \leq m(n, z) + K_z(n'') < m(n, z) + K_z(\overline{n}(z)) .$$

Thus, the firm chooses the target whenever it can finance it internally and distributes any additional cash. Since the target is affordable at $\widetilde{n}$, the same conclusion holds for every $n \geq \widetilde{n}$. ■

Figure 1 illustrates both parts of the proposition. Panel (a), drawn at the lowest productivity state, is the firm-side counterpart of Figure I in Aiyagari (1994).[20] Next-period cash rises with cash on hand while the firm cannot finance its target internally and becomes flat once the target is affordable.[21] Net payout is zero at the target-financing threshold; above it, each additional unit of cash on hand is distributed one-for-one.

Panel (b) displays the target at four productivity states. The flat segment of each policy identifies $\overline{n}(z)$, whose height varies markedly with productivity. The arrows indicate the leftward/downward shift of the policy function as productivity rises, not merely the decline in $\overline{n}(z)$. Panel (b) also admits a dynamic interpretation: for a firm with productivity z, the policy maps current cash into next-period cash. Iterating this mapping, cash declines where the curve lies below the 45-degree line and accumulates where it lies above; a crossing therefore identifies a policy fixed point. When the flat segment crosses the 45-degree line, its height $\overline{n}(z)$ is the fixed point. Otherwise, $\overline{n}(z)$ remains an upper bound, and the local dynamics are governed by the sloped portion of the policy. Proposition 3 explains the flat segment; we next establish how productivity orders the targets.[22]

[20] Net payout d corresponds to consumption c, and next-period cash holdings n' correspond to next-period wealth a_{t+1}.

[21] This resembles the buffer-stock saving model of Carroll (1997). Although g is continuously differentiable, its second derivative changes at $d = 0$, so the cash policy may kink at the boundary between equity issuance and positive net payout.

[22] It is worth noting that the implication of Proposition 3 differs from Proposition 4 in Aiyagari (1993), which suggests that households with excessively high wealth gradually decumulate. In contrast, the target cash result here implies that a firm with an excessively large cash stock immediately reduces it to the target level.

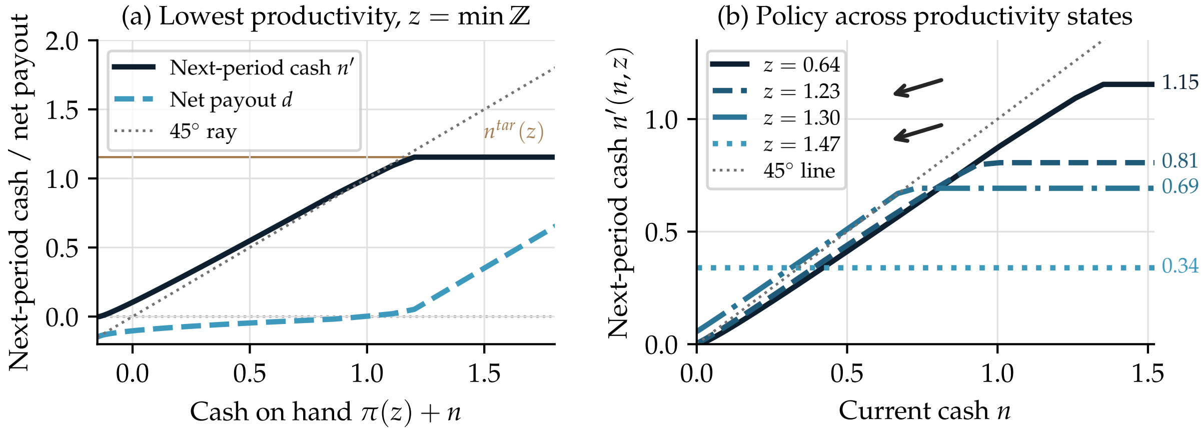

Figure 1: Cash-holding and net-payout policy functions

Notes: Panel (a) shows cash policy and net payout at the lowest productivity; panel (b) shows policy functions across productivity. Arrows indicate the policy-function shift with productivity; the flat target is a fixed point only when the policy intersects the 45-degree line.

4.2 Cash policies across productivity states

Persistent increases in productivity lower the firm's target cash holdings—a global comparative static. The cash policy itself, however, need not be monotone in productivity: for a firm issuing equity, higher productivity relaxes its current financing need and can initially raise cash accumulation, even as the target declines. This second result is local, and it accounts for the hump-shaped policies in the calibrated economy.

The first result rests on two conditions. First, because $\rho_z > 0$, our AR(1) productivity process implies that higher current productivity shifts the distribution of z' upward in the sense of first-order stochastic dominance (FOSD). Second, cash has a decreasing precautionary value. Rather than assuming the second property, we derive it from the model's primitives.

Proposition 4 (The target falls with productivity)**.** *Under the conditions of Proposition 3, suppose the transition kernel of idiosyncratic productivity is weakly increasing in the sense of FOSD and operating profits $\pi(z)$ are weakly increasing. Then, $\overline{n}(z)$ is weakly decreasing in z.*

Proof. We first establish the required ordering of the continuation value. Under the maintained concavity of g, the weak monotonicity of $\pi(z)$, and the FOSD ordering of the productivity transition kernel, the steady-state Bellman operator preserves functions that are increasing and concave in cash and have decreasing differences in (n, z). Because the operator is a β-contraction, its fixed point J inherits these properties. Consequently, for

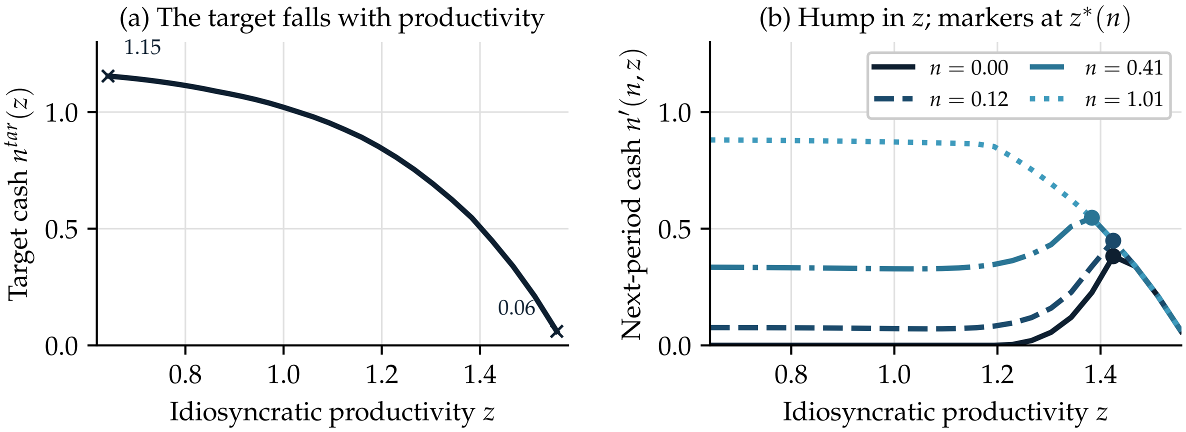

Figure 2: The falling target and the cross-sectional cash policy

Notes: Panel (a) plots the target cash level $\overline{n}(z)$ against idiosyncratic productivity; it is strictly decreasing, from 1.15 to 0.06. Panel (b) plots next-period cash $n'(n,z)$ against productivity for four initial cash positions, with markers at the interior maximizer $z^*(n)$. For low initial cash the policy is hump-shaped; for high initial cash the firm is above its target throughout most of the productivity range and decumulates, so no interior peak exists.

every $n'_2 > n'_1$,

$$J(n'_2, z') - J(n'_1, z')$$

is weakly decreasing in z'. Appendix C provides the complete Bellman-operator invariance argument.

Fix $n'_2 > n'_1$. The difference in the target objective is

$$K_z(n'_2) - K_z(n'_1) = -q^n(n'_2 - n'_1) + \beta \mathbb{E}[J(n'_2, z') - J(n'_1, z') \mid z].$$

The expression inside the expectation is weakly decreasing in z', while the maintained transition kernel shifts upward with current z. The expected gain from choosing n'_2 rather than n'_1 is therefore weakly decreasing in z. Hence, $K_z(n')$ has decreasing differences in (n', z). Monotone comparative statics and the uniqueness of the maximizer then imply that $\overline{n}(z)$ is weakly decreasing in z. ■

The decreasing-differences property formalizes the precautionary mechanism: an additional unit of cash is less valuable when the firm is more productive because higher expected operating profits reduce its exposure to costly equity issuance. Panel (a) of Figure 2 confirms this comparative static in the calibrated economy. The target declines throughout the productivity grid, so the most productive firms are effectively satiated at the cash positions they typically hold. Panel (b) shows how the declining target shapes the cross-sectional cash policy. For firms entering with low cash holdings, next-period cash first rises with productivity and then falls as the target declines. Firms entering with ample cash are already above their target over much of the productivity range and decumulate toward it. As initial cash rises, the peak shifts toward lower productivity and eventually disappears.

The hump reflects the interaction of two opposing forces. First, the precautionary target falls with productivity because persistent high revenues reduce expected financing needs. Second, internal cash flow $\pi(z)$ rises with productivity and expands the firm's capacity to accumulate. Low-productivity firms therefore have a high precautionary target but insufficient internal cash flow to reach it, whereas high-productivity firms have ample internal cash flow but little precautionary need. A hump can emerge at intermediate productivity, where profits have become sufficient to support accumulation but the target has not yet fallen too far. To isolate these forces locally, we consider the cash Euler equation at an interior choice:

$$q^n g'(d) = \beta \mathbb{E}[J_n(n', z') \mid z], \qquad d = \pi(z) + n - q^n n'. \tag{6}$$

The next proposition characterizes how productivity affects next-period cash at an interior point. It is current equity issuance—not merely exposure to possible issuance in future states—that activates the current cash-flow channel. Future issuance risk is already summarized by the expected marginal continuation value of cash $\beta \mathbb{E}[J_n(n', z') \mid z]$ in every region. When $d < 0$, higher productivity reduces the marginal cost of the equity the firm is raising, but it also lowers the precautionary value of cash by improving expected future financing conditions. Once $d > 0$, current payout is locally linear, so operating profits drop out of the marginal condition and only the second force remains.

Proposition 5 (When next-period cash rises with productivity)**.** *Denote the expected marginal continuation value of cash by*

$$M(n', z) \equiv \beta \mathbb{E}[J_n(n', z') \mid z].$$

Suppose π and M are continuously differentiable and the cash choice is interior, with $d \neq 0$ and $M_{n'}(n', z) < 0$. Then the policy is locally differentiable and

$$\frac{\partial n'}{\partial z} = \frac{M_z(n', z) - q^n g''(d) \pi_z(z)}{-(q^n)^2 g''(d) - M_{n'}(n', z)}. \tag{7}$$

The denominator is positive. If $d < 0$, then $g''(d) = -\mu$, and next-period cash rises with produc-

tivity if and only if

$$q^n \mu \pi_z(z) > -M_z(n', z).$$

If $d > 0$, then $g''(d) = 0$, so next-period cash has the sign of $M_z(n', z)$ and therefore weakly decreases when $M_z(n', z) \leq 0$. At $d = 0$, the change in the curvature of g can generate a kink, so the formula applies separately on either side of the payout boundary.

Proof. Differentiate (6) with respect to z, using

$$\frac{\partial d}{\partial z} = \pi_z(z) - q^n \frac{\partial n'}{\partial z}.$$

Collecting terms in $\partial n'/\partial z$ gives (7).[23] Concavity implies $g'' \leq 0$, and $M_{n'} < 0$, so the denominator is positive. Substituting $g'' = -\mu$ in the issuance region yields the stated inequality. When $d > 0$, $g'' = 0$, so the sign of the policy response equals the sign of M_z. ■

Wherever the derivative exists, the decreasing-differences result established in the proof of Proposition 4 implies $M_z(n', z) \leq 0$. In the positive-payout region, where $g''(d) = 0$, Proposition 5 therefore implies $\partial n'/\partial z \leq 0$. Any smooth interior branch along which cash rises with productivity must consequently lie in the equity-issuance region. Panel (b) of Figure 2 displays such a branch: holding current cash fixed, next-period cash initially increases with productivity.

In the issuance region, higher productivity generates two opposing forces. By reducing the amount of equity that the firm must raise, it provides issuance relief of $q^n \mu \pi_z(z)$, encouraging cash accumulation. At the same time, it lowers the precautionary continuation value of cash by $-M_z(n', z)$. Next-period cash rises when the first force is stronger and falls when the second dominates. In the calibrated model, issuance relief dominates initially and the precautionary-value effect dominates later, producing the hump. That these forces cross exactly once is a quantitative property of the calibration, not a global theoretical result.

Our mechanism differs from the standard household saving mechanism in Aiyagari (1994) and Krusell and Smith (1998), where the borrowing constraint is the central source of nonlinearity. Here, an additional nonlinearity arises away from the constraint because firms become satiated at targets that vary with persistent productivity. Aggregate shocks can therefore move firms among distinct financing, active-accumulation, and satiated regions.

[23] For this local characterization, we consider the continuous-state productivity process underlying its finite-state computational approximation.

4.3 The marginal propensity to pay out across the firm distribution

The marginal propensity to pay out, introduced in Definition 1, is

$$MPPO(n,z) = \frac{\partial d(n,z)}{\partial n} = 1 - q^n \frac{\partial n'(n,z)}{\partial n}. \tag{8}$$

The MPPO differentiates payout with respect to current cash, so the response of n' to productivity does not by itself determine $\partial n'/\partial n$. We therefore derive the local result along the cash dimension and then use panel (b) of Figure 3 to show how the resulting MPPO varies across productivity in the calibrated economy—the cross-sectional perspective that connects to the profitability-based empirical evidence.

Proposition 6 (The MPPO at an interior cash choice)**.** *Suppose the cash choice is interior and differentiable at a point with $d \neq 0$ and $M_{n'}(n',z) < 0$. Define*

$$A(d) \equiv -g''(d) \geq 0, \qquad B(n',z) \equiv -M_{n'}(n',z) > 0.$$

Then

$$\frac{\partial n'(n,z)}{\partial n} = \frac{q^n A(d)}{(q^n)^2 A(d) + B(n',z)}, \tag{9}$$

$$MPPO(n,z) = \frac{B(n',z)}{(q^n)^2 A(d) + B(n',z)} \in (0,1]. \tag{10}$$

If $d > 0$, then $A(d) = 0$ and $MPPO = 1$. If $d < 0$, then $A(d) = \mu$ and $MPPO \in (0,1)$. At $d = 0$, the cash policy need not be differentiable; the formulas instead characterize the one-sided derivatives. The MPPO approaches a value below one from the issuance region and equals one on the positive-payout side.

Proof. Differentiate (6) with respect to current cash n, holding z fixed:

$$q^n g''(d)\left(1 - q^n \frac{\partial n'}{\partial n}\right) = M_{n'}(n',z)\frac{\partial n'}{\partial n}.$$

Solving for $\partial n'/\partial n$ and substituting into (8) gives (10). ■

Holding productivity fixed, Proposition 6 asks how the firm allocates one additional unit of current cash between net payout and next-period cash holdings. The two curvature terms have a direct interpretation: $A(d)$ measures how strongly the marginal equity issuance cost changes with current net payout, while $B(n',z)$ measures how quickly the marginal continuation value falls as the firm carries more cash forward.

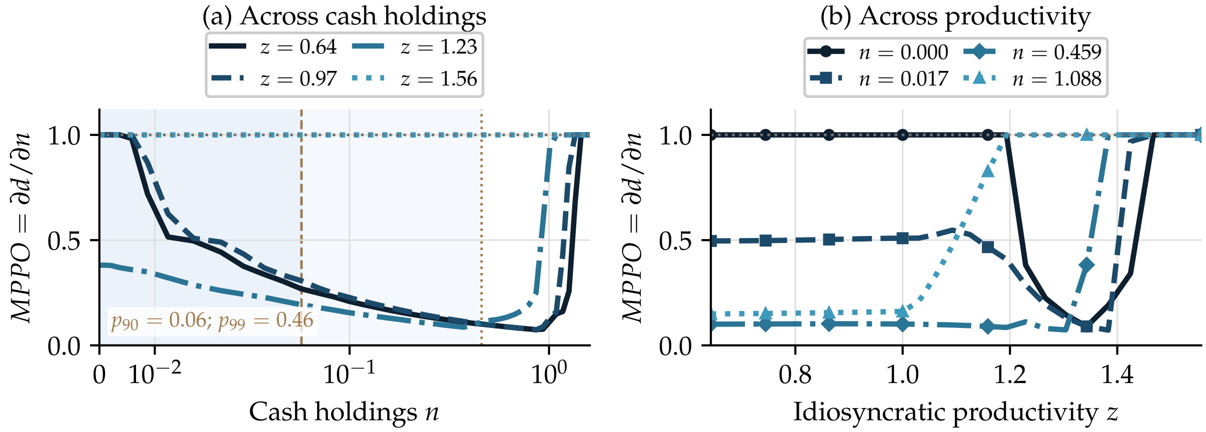

Figure 3: The MPPO across cash holdings and productivity

Notes: Panel (a) plots $MPPO(n,z) = 1 - q^n \partial n'(n,z)/\partial n$ against cash holdings at four productivity states. The horizontal axis is on a symmetric-log scale because the stationary distribution is concentrated near the origin. The darker shaded region extends to the 90th percentile of stationary cash holdings, $n = 0.06$; the lighter extension reaches the 99th percentile, $n = 0.46$. For the three lower productivity states, the schedule is U-shaped: it equals one at the constraint, falls while the firm accumulates, and returns to one at the target. At $z = 1.56$, the firm is satiated throughout the plotted range and the MPPO is one. Panel (b) plots the MPPO against idiosyncratic productivity at four current-cash positions.

With a positive net payout, the current payoff is locally linear and an additional unit of cash is distributed in full. During equity issuance, the convex financing cost gives the firm an incentive to retain part of that unit, so the MPPO lies below one. This is an interior result and does not by itself describe the boundary regions or imply that the MPPO is globally monotone in cash.

We complete the policy map with two boundary cases. If the no-borrowing constraint $n' = 0$ binds strictly, the cash policy is locally constant and $MPPO = 1$. In the flat target region of Proposition 3, $n' = \overline{n}(z)$ is again independent of current cash, so $MPPO = 1$. Between these boundaries, an issuing firm retains part of a marginal dollar and has $MPPO < 1$. At a fixed productivity, the MPPO therefore follows a high–low–high pattern across current cash holdings: full pass-through at the constraint, partial retention during active accumulation, and full pass-through again at satiation. In the calibrated economy, this pattern produces the U-shape in Figure 3.

Panel (a) of Figure 3 displays this schedule in the calibrated economy. At lower productivity, the MPPO falls as the firm accumulates cash and returns to one at the target. As productivity rises, the target falls and the accumulation region narrows. When the target is close to zero, the firm is satiated throughout the relevant range of cash holdings and the MPPO remains at one.

Panel (b) provides the complementary cross-sectional view. At a fixed cash position, productivity moves firms among the constrained, active-accumulation, and satiated regions. The resulting MPPO is most clearly U-shaped at low cash: pass-through is high at low and high productivity and lower at intermediate productivity. The exact shape shifts with the firm's cash position. The two arms of the U have the same MPPO for opposite economic reasons. At the constraint, the firm passes marginal resources through because it *cannot* accumulate cash: an additional unit of cash reduces equity issuance. At the target, it passes resources through because it *need not* accumulate cash: its precautionary buffer is already full. A sufficiently large shock can nevertheless move a satiated firm back into the active-accumulation region, where it begins to retain.

Note that the shaded bands in panel (a) show that the stationary distribution is concentrated near the left arm of the U: most firms are constrained or actively accumulating, while few are at their target. The calibrated distribution places little mass on the declining branch of the cash policy in productivity. Over the empirically populated region, the model therefore predicts that the MPPO decreases with profitability — equivalently, firms retain a larger share of marginal internal resources. The reversal on the sparsely populated branch is quantitatively less relevant. We take this prediction to the data in Section 5.2.

Since the MPPO differs across these regions, aggregate payout behavior depends on how the distribution allocates firms among them rather than on any single firm's position. Restoring the aggregate state X, we summarize that allocation by the average MPPO,

$$\overline{MPPO}(X) = \int MPPO(n, z; X)\, d\Phi(n, z), \tag{11}$$

which averages firm-level MPPOs over the current joint distribution Φ. Aggregate shocks shift Φ across the financing, active-accumulation, and satiated regions, and $\overline{MPPO}(X)$ moves with the distribution. This statistic carries the firm-level policies characterized here into the state-dependent payout and consumption responses of Section 5, where we track it against aggregate liquidity over the simulated business cycle.

5 Corporate liquidity and aggregate fluctuations

Corporate liquidity shapes aggregate fluctuations through two linked margins. The contemporaneous firm distribution determines how strongly payouts respond to an aggregate shock; the payout response determines how much of that shock reaches household spending. Our global nonlinear solution brings these margins together in general equilibrium: heterogeneous firms move across financing, active-accumulation, and satiation regions, allowing the nonlinear firm-level liquidity policies characterized above to generate state-dependent aggregate consumption responses. We first describe the calibration and solution and validate both links with firm- and household-level evidence. We then present the central state-dependence result, connect it to the distribution-weighted MPPO, quantify the role of firm heterogeneity, and apply the same retention mechanism to transfers to firms.

5.1 Solution and calibration

Solution method Despite its relatively parsimonious formulation, our model's computation of recursive competitive equilibrium presents several significant methodological challenges: 1) non-linear aggregate fluctuations; 2) non-trivial market clearing conditions for both labor and consumption goods; and 3) occasionally binding constraints. We employ the global non-linear solution method developed by Lee (2026a), which can efficiently solve this problem. The method solves the problem backward over a long sequence of simulated exogenous aggregate states. During this backward iteration process, conditional expectations are calculated by integrating the realized value functions from previous *iterations*, thereby eliminating the need to explicitly specify aggregate laws of motion. For more details on the method and a performance comparison with the Krusell and Smith (1998) algorithm, please see the Appendix D.

For tractability, we normalize the firm's value function by contemporaneous consumption c_t following Khan and Thomas (2008). We define the marginal utility price of consumption as $p_t := 1/c_t$ and the normalized value function $\tilde{J}_t := p_t J_t$. From the household's intratemporal and intertemporal optimality conditions, we have $w_t = \eta (l_t^H)^{1/\chi}/p_t$ and $q_t = \beta \mathbb{E}[p_{t+1}]/p_t$. Thus, p_t, together with the wage implied by labor-market clearing, characterizes the equilibrium price system.

Calibration The model period is a quarter. We set the household discount factor to $\beta = 0.995$, implying an annualized real rate of approximately 2% at the non-stochastic steady state, and the Frisch labor-supply elasticity to one. On the firm side, the span-of-control parameter is 0.930, and the persistence and volatility of idiosyncratic productivity follow Bachmann et al. (2013). The net return on corporate liquidity is 87% of the risk-free rate, following Cooley and Quadrini (2001). We take the limiting case $\zeta \to 0^+$, so external suppliers accommodate firms' aggregate demand at the cash price implied by the return wedge. Hence, the quantity of liquid assets is determined by market clearing and no

Table 1: Fixed Parameters

Parameter	Description	Value	Source
Households			
β	Discount factor	0.995	Chen (2017)
χ	Frisch labor supply elasticity	1.000	Kaplan et al. (2018)
Production			
γ	Span of control	0.930	Standard calibration
λ	Cash return wedge	0.870	Cooley and Quadrini (2001)
ρ_z	Idiosyncratic shock persistence	0.861	Bachmann et al. (2013)
σ_z	Idiosyncratic shock volatility	0.075	Bachmann et al. (2013)
Aggregate			
$p(A_B \mid A_B)$	Persistence of low aggregate TFP	0.875	Krusell and Smith (1998)
$p(A_G \mid A_G)$	Persistence of high aggregate TFP	0.875	Krusell and Smith (1998)

Notes: This table presents the fixed parameters used in the model, along with their sources. For the cash-return wedge, we back out the ratio wedge λ implied by the calibration used in Cooley and Quadrini (2001): $r/(1/\beta - 1)$. For the idiosyncratic shock volatility σ_z we sum both the idiosyncratic and sector-specific shock volatilities from Bachmann et al. (2013).

separate liquid-asset price must be solved for. Table 1 summarizes the fixed parameters.

We calibrate the remaining parameters to aggregate moments. The external-financing-cost parameter μ is identified by the ratio of aggregate corporate liquidity to output: a larger μ strengthens the precautionary motive and raises firms' desired liquidity. The operating cost ξ is identified primarily by the consumption-to-output ratio, while the labor-disutility parameter η targets hours and the aggregate TFP spread Δ_A targets output volatility. The data moments are quarterly averages over 1970–2019.[24]

Table 2 summarizes the resulting calibration and model fit. The first panel pairs each calibrated parameter with its principal target and reports the corresponding data and model moments. The second panel shows that the model also generates an untargeted dividend-to-output ratio of 1.98%, compared with 1.91% in the data. We turn to the untargeted firm-level moments in the bottom panel next.

[24] For now, we set lump-sum taxes on households and transfers to firms to zero. Section 5.4 describes their calibration for the fiscal-policy exercises.

Table 2: Calibrated Parameters and Target Moments

Parameters	Description	Data	Model	Calibration
Targeted moments				
μ	Corporate cash holdings relative to output (%)	10.00	9.95	0.08
ξ	Consumption relative to output (%)	66.00	63.63	0.15
η	Hours worked relative to time available (%)	33.00	32.90	13.50
Δ_A	Output volatility (% *p.q.*)	1.45	1.72	0.02
Untargeted aggregate moments				
	Dividend to output (%)	1.91	1.98	
Untargeted micro moments				
	$\mathrm{Corr}(n/y, y)$	-0.293	-0.078	
	$\mathrm{Corr}(\mathrm{std}(n/y)_p, \bar{y}_p)$	-0.972	-0.393	
	Mitigation of dividend drop from 1 s.d. higher liquidity	50%	50%	
	$MPPO_{Q1} - MPPO_{Q4}$	0.42	0.40	

Notes: This table presents the calibrated parameters along with the corresponding target moments, observed data values, model-implied values and the respective parameter values in the last column. The observed targeted data values are averages over quarterly data spanning from 1970 to 2019. In addition, the table reports the model fit to aggregate and micro-level moments. For the first two micro moments, n/y denotes cash-to-sales in Compustat and cash-to-output in the model; y is real sales in Compustat and output in the model. The second row computes the standard deviation of residual n/y within each sales/output bin p and correlates it with average bin scale, $\bar{y}_p$. The third row is $100 \times \gamma_h \sigma_n / |\beta_h|$, where β_h is the dividend response to a negative aggregate-growth year, γ_h is the interaction with lagged liquidity and σ_n is the standard deviation of liquidity. The last row is the impact MPPO gap between the lowest and highest lagged-profit quartiles in Compustat and the corresponding output-state quartiles in the model.

5.2 Micro evidence on the transmission mechanism

We next evaluate whether the calibrated model reproduces micro moments that are central to the transmission mechanism but are not directly targeted. Firm-level data discipline the liquidity distribution and payout response, while a PSID exercise assesses whether payout income reaches household spending. Table 2 summarizes the firm-side fit across four untargeted moments; Appendix A.4 provides the underlying figures and additional implementation details, and Appendix A.2 describes the Compustat sample construction and cleaning.

Firm liquidity and payout behavior The first row captures the average relation between liquidity and firm scale. In Compustat, we compute cash-to-sales using cash and short-term investments divided by real sales, and correlate this ratio with real sales. In the model, we compute the analogous correlation between cash-to-output and output. The second row uses the dispersion of liquidity. We first residualize cash-to-sales in the data using lagged log real assets, leverage, real sales growth, firm fixed effects, and sector-year fixed effects. We then sort firms into sales percentiles, compute the standard deviation of residual cash-to-sales within each percentile, and correlate that dispersion with average sales. The model counterpart computes the same object across output/productivity bins. Thus, sales is the scale variable for the cash-distribution moments; profits enter only in the MPPO exercise below, where we need a proxy for the firm's profitability state. The model reproduces the negative signs of both relationships, although smaller in magnitude: the model correlations are -0.078 and -0.393, compared with -0.293 and -0.972 in the data.

The third row summarizes the dividend-smoothing exercise. In Compustat, we estimate local projections of log real dividends on an indicator for years in which real GDP growth is negative, lagged cash/assets, their interaction, lagged size, leverage, sales growth, and firm fixed effects. If β_h is the dividend response in a negative-growth year and γ_h is the interaction with lagged cash-to-assets ratio, the reported number is $100 \times \gamma_h \sigma_n / |\beta_h|$. Hence, 50% means that a one-standard-deviation increase in lagged cash-to-assets ratio reduces the implied dividend cut by about one half. The model entry is computed from the same regression on simulated data, replacing cash-to-assets ratio with cash-to-output ratio, considering only firms with weakly positive dividends and defining downturns as low aggregate TFP states. Panel (a) of Figure 4 plots the horizon-specific interaction coefficients γ_h in the data and the model, showing that firms with more liquidity cut dividends less during downturns; the table summarizes this pattern as a roughly 50% mitigation for a one-standard-deviation increase in liquidity, while the panel reports the underlying coefficients rather than this standardized statistic.[25]

The last row compares the model MPPO to an empirically estimated payout pass-through from temporary expansions of U.S. net operating loss (NOL) carrybacks. A carryback converts a current tax loss into a cash refund by applying it against past taxable income, making these reforms a source of policy-induced liquidity variation. Under the pre-existing rule, a firm with a taxable loss could carry the loss back two years and claim a

[25]Papers such as Opler et al. (1999) and Gao et al. (2013) show that large, publicly listed firms tend to hold more cash, and that these firms use this cash to smooth their dividend over time, in line with our results.

refund of federal income taxes paid in those two years. The Job Creation and Worker Assistance Act of 2002 expanded the carryback window from two to five years for business losses around the 2001 recession. The Worker, Homeownership, and Business Assistance Act of 2009 again allowed firms to use a five-year carryback for one loss year, either 2008 or 2009. Thus, the policies did not provide a uniform transfer to all firms. They increased liquidity only for loss-making firms that had paid enough federal taxes in years that were outside the old two-year window but inside the new five-year window.

This policy variation is useful for identification because the extra refund is mechanically determined by current NOLs and taxes paid before the reform-relevant payout decision. Conditional on the size of the current loss and the refund capacity already available under the old two-year rule, the incremental refund comes from taxes paid in years three to five before the loss. Those tax payments are predetermined, and they became valuable only because Congress unexpectedly expanded the carryback window during downturns. We therefore interpret the incremental predicted refund as a policy-induced liquidity shock rather than as an endogenous payout choice.

The empirical shock is the additional refund a firm can claim under the new rule relative to the old rule.

For each eligible loss year ℓ, we compute

$$\widehat{Refund}_{i\ell} = \min\{0.35 NOL_{i\ell}, Capacity_{i\ell}^{new}\} - \min\{0.35 NOL_{i\ell}, Capacity_{i\ell}^{old}\}.$$

The old capacity is federal taxes paid in the previous two years, while the new capacity is taxes recoverable under the expanded five-year window. We multiply the NOL by 0.35, the top federal corporate income tax rate prevailing during both reform episodes, to approximate the tax value of the loss. The minimum operator appears because the refund cannot exceed either the tax value of current losses or the taxes previously paid in eligible carryback years. For 2002, we sum the predicted refunds from 2001 and 2002 losses and assign them to the 2002 payout event, because the reform was enacted early in 2002 and firms could receive or anticipate refunds associated with both loss years. For 2009, firms could elect the expanded carryback for either 2008 or 2009 losses, but not both, so we assign the larger predicted refund to the 2009 payout event.

We regress total payout over lagged assets on the predicted refund over lagged assets interacted with quartiles of lagged operating profits, controlling for lagged size, leverage, sales growth, the tax value of current losses, old-rule refund capacity, and sector-year fixed effects. Total payout is dividends plus repurchases minus equity issuance. Each coefficient is dollars of payout per dollar of predicted liquidity in a profit quartile.

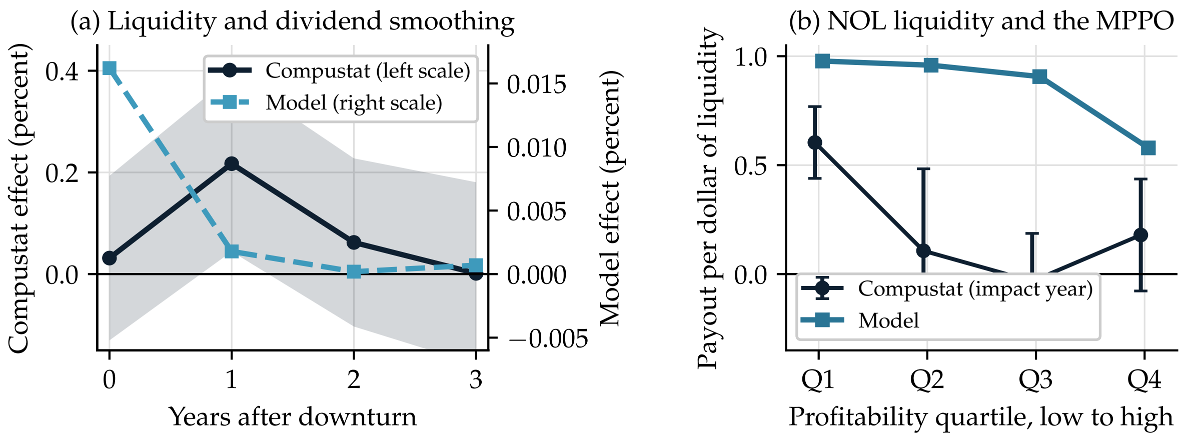

Figure 4: Firm liquidity and payout pass-through

Notes: Panel (a) plots the horizon-specific interaction between a downturn and lagged liquidity in Compustat (left scale, with 90% confidence intervals) and the model (right scale); these coefficients are distinct from the standardized mitigation statistic in Table 2. Panel (b) compares the impact-year response of net payout to a predicted NOL refund across lagged-profit quartiles, with 90% confidence intervals, to the model-implied average MPPO across output quartiles.

Panel (b) of Figure 4 shows that net-payout pass-through is highest in the lowest-profitability quartile and substantially lower in the remaining quartiles. The table reports $Q_1 - Q_4$: in the data, $0.604 - 0.180 = 0.424$ with $p = 0.021$; in the model, the distribution-weighted output-quartile gap is $0.978 - 0.579 = 0.399$. The model therefore matches the sign and magnitude of the key micro facts: liquidity is higher and more dispersed among smaller firms, cash smooths dividends in downturns, and liquidity shocks pass through more strongly to payouts among low-profit firms.[26] The comparison disciplines the shape and approximate magnitude of payout pass-through over the empirically populated part of the distribution; it does not require each empirical quartile to map exactly into a theoretical policy regime. Appendix A.4 reports reform-specific estimates, coefficient-equality tests, implementation details, and the pre-event placebo.

Dividend income and household expenditure The second link requires changes in corporate payouts to reach household spending. Existing evidence shows that consumption responds positively to dividends, and by more than to capital gains (Baker et al., 2007; Di Maggio et al., 2020; Bräuer et al., 2022). We complement this cross-sectional evidence with a within-household analysis using the Panel Study of Income Dynamics (PSID). The

[26] Although the model implies a U-shaped MPPO in theory, the quantitative calibration places only the top 2% of firms on the upward-sloping part of the schedule. Over the empirically relevant range, the model therefore predicts that the MPPO declines with firm profitability. Consistent with this prediction, our quartile-based estimates show a decreasing MPPO profile across the profitability distribution.

Table 3: Effects of real dividend income on real expenditure

	Real Expenditure	Real Expenditure	Real Expenditure	Real Expenditure
Real dividend income	0.157***	0.0886**	0.0814*	0.0785*
	(0.0338)	(0.0416)	(0.0428)	(0.0412)
Household controls		✓	✓	✓
Household fixed effects			✓	✓
Year fixed effects				✓
Observations	46 419	23 138	21 223	21 223

Notes: This table reports the effect of a one-dollar increase in real dividend income on real expenditure. Household covariates include financial and business wealth, housing wealth, financial asset income, labor income, age, and education. Standard errors are clustered at the household level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

PSID provides biennial observations from 2005 to 2021 on household dividend income and total expenditure.[27]

We estimate

$$c_{it} = \beta d_{it-1} + \Gamma X_{it-1} + \mu_i + \mu_t + \epsilon_{it}, \tag{12}$$

where c_{it} is household expenditure and d_{it-1} is lagged dividend income. The vector X_{it-1} contains lagged labor income, financial income, non-housing wealth, housing wealth, age, and education. Household fixed effects μ_i absorb time-invariant household characteristics, while year fixed effects μ_t absorb aggregate shocks common to all households. The coefficient β therefore measures whether within-household changes in dividend income are associated with subsequent changes in expenditure, conditional on other income and wealth.

Table 3 shows a positive association across all specifications. In the preferred specification, which includes household covariates and household and year fixed effects, a one-dollar increase in dividend income is associated with a 7.85-cent increase in real expenditure. The estimate is consistent with earlier evidence from the Consumer Expenditure Survey and transaction data (Baker et al., 2007; Di Maggio et al., 2020; Bräuer et al., 2022). Together with the firm-side results, it supports the complete transmission mechanism: liquidity affects corporate payouts, and variation in payout income passes through to household spending. Having validated both links in the micro data, we now use the model to trace how this empirically disciplined payout channel generates state-dependent aggregate dynamics.

[27] Appendix A.3 describes the construction of the PSID panel. Total expenditure includes food, housing, health care, transportation, childcare and education, clothing, travel, and recreation.

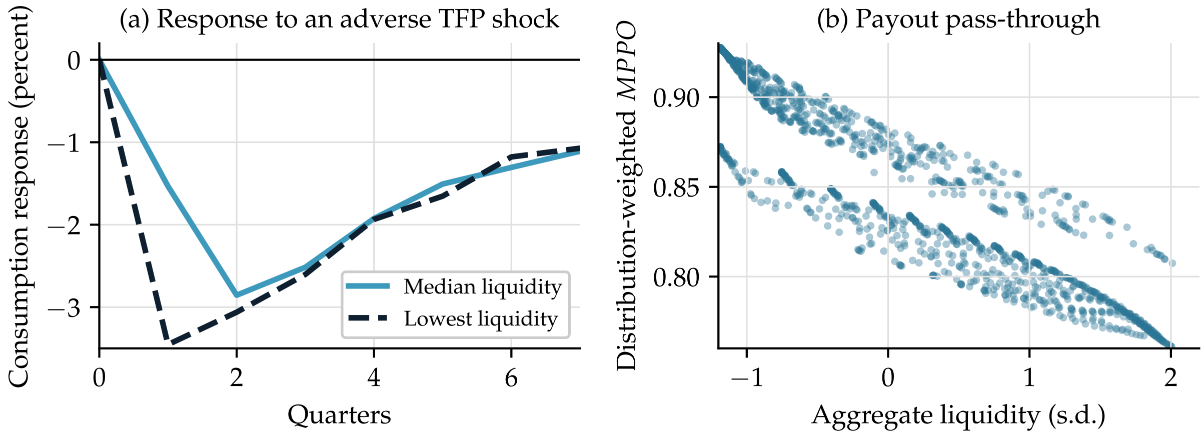

Figure 5: Payout pass-through and consumption responses across liquidity states

Notes: Panel (a) plots the generalized impulse response of consumption to the same adverse aggregate TFP shock from the median-liquidity distribution (light-blue solid line) and the lowest-liquidity distribution observed in the simulation (dark-blue dashed line). The latter is roughly two standard deviations below mean liquidity and about 80% below the median. Panel (b) plots the distribution-weighted MPPO against lagged aggregate liquidity over the simulated business cycle. A one-standard-deviation increase in aggregate liquidity is associated with a 4.67 percentage-point decline in the average MPPO.

5.3 State-dependent payout and consumption responses

This section explores the role of corporate cash in shaping consumption dynamics. In the model, aggregate productivity A_t switches between two states, A_G and A_B, following a persistent Markov process. We define a negative shock as a transition from A_G to A_B and a positive shock as the reverse.

A key mechanism in the model is that firm-level payout sensitivity to a TFP shock depends on cash holdings. Section 4.3 shows that the MPPO exhibits a U-shaped relationship with cash. Firms operating at the cash constraint respond aggressively to negative shocks by cutting payouts, while cash-rich firms at the target also display an MPPO of one. This generates state-dependent household consumption responses, as consumption depends on net payouts and therefore on the aggregate cash stock, as established in Section 2.

Figure 5 summarizes the central quantitative result and the mechanism behind it. Panel (a) reports the generalized impulse response function (GIRF) of consumption to a negative TFP shock at different levels of cash. When cash is at its lowest level in the simulation, roughly two standard deviations below its mean and about 80% below the median, the consumption drop is almost three times as large as when cash is at its median level. This amplification reflects the high MPPO in low-cash states, which forces firms to pass the shock through to households almost one-for-one.

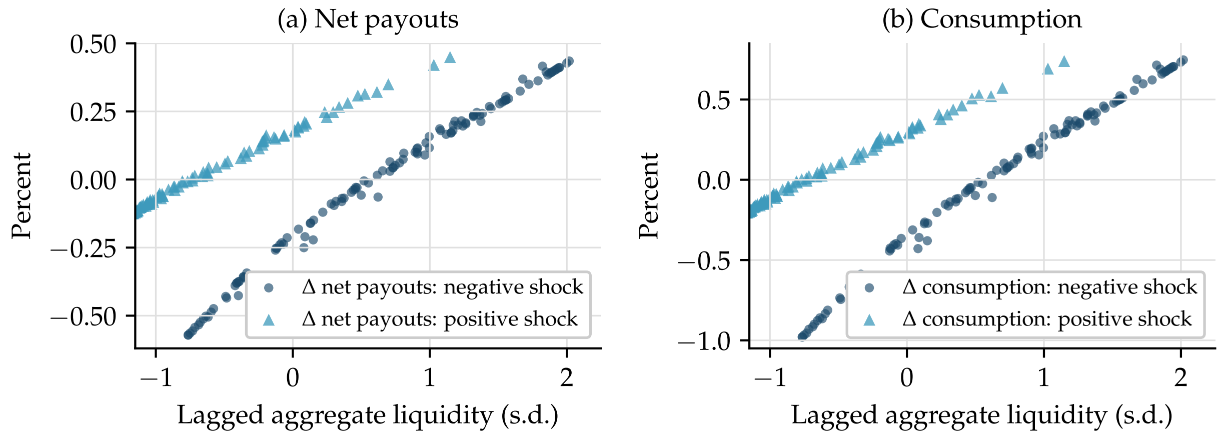

Figure 6: Endogenous state dependence in the shock responses of net payouts and consumption

Notes: The figure plots net-payout and consumption responses to negative (dark-blue circles) and positive (light-blue triangles) aggregate TFP shocks against lagged aggregate liquidity. Responses are normalized by their shock-specific average. Panel (a) reports net payouts and Panel (b) consumption. The TFP innovation has the same magnitude in every observation.

Panel (b) relates this amplification to firms' payout behavior. It plots the distribution-weighted MPPO defined in (11) over the business cycle as a function of aggregate cash holdings. The figure shows a clear negative relationship: a one-standard-deviation increase in cash reduces the MPPO by 4.67 percentage points. While the theoretical results in Sections 2 and 4.3 imply a U-shaped relationship at the firm level, the aggregate cash levels observed over the business cycle lie predominantly on the left side of the U, where the MPPO declines with cash. Despite the negative correlation between cash and the MPPO, there remains dispersion in the MPPO for similar levels of cash. This dispersion is driven mainly by the aggregate productivity state: when TFP is high, more firms are pushed away from the constraint, lowering the aggregate MPPO.

Panel (a) of Figure 5 compares two initial distributions and one adverse shock. Figure 6 shows that the same mechanism operates systematically across all realized transitions and for both shock directions. It plots the responsiveness of net payouts and consumption to negative (dark-blue circles) and positive (light-blue triangles) aggregate TFP shocks as a function of lagged aggregate cash holdings. The magnitude of the aggregate shock is fixed at $|A_G - A_B| = 4\%$. Hence, variation in consumption responses across periods reflects endogenous state dependence rather than differences in shock size.

For negative aggregate shocks, consumption responsiveness decreases with aggregate cash holdings: when firms are liquid, they smooth payouts and shield households from income losses. For positive aggregate shocks, consumption responsiveness increases with

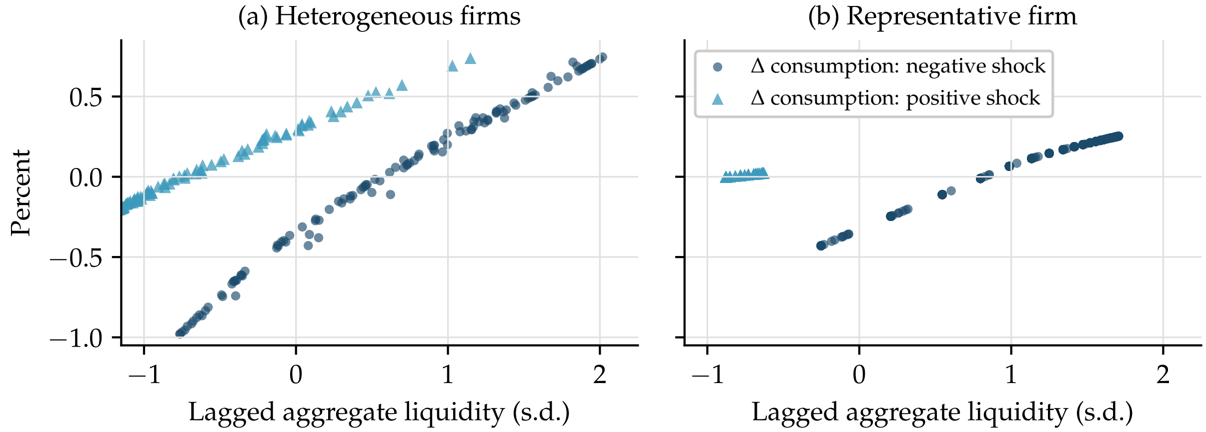

Figure 7: Endogenous state dependence in consumption responses to negative and positive shocks in the baseline (a) and representative-firm (b) models

Notes: The figure plots consumption responses (relative to average consumption, in percentages) to negative and positive aggregate TFP shocks as a function of lagged aggregate liquidity. Panel (a) plots the baseline heterogeneous-firm model, while Panel (b) plots the representative-firm model.

aggregate cash, reflecting a greater pass-through of productivity gains to households. This asymmetric pattern reflects firms' precautionary behavior and is mirrored in the net-payout responses shown in the left panel of Figure 6. Quantitatively, a one-standard-deviation increase in lagged aggregate cash reduces the consumption decline in response to a negative TFP shock by 0.61 percentage points. For a positive TFP shock, the corresponding change is 0.42 percentage points.

Aggregate cash holdings therefore provide a consumption buffer against negative aggregate shocks by smoothing payouts. At the same time, higher cash holdings facilitate the pass-through of positive productivity shocks to consumption. The buffering effect on the downside is quantitatively stronger than the amplification effect on the upside.

Role of heterogeneity To illustrate the importance of firm heterogeneity in driving consumption state dependence, we compare the baseline model with a representative-firm model. We shut down the idiosyncratic productivity shock while retaining market incompleteness and aggregate productivity shocks. At the same time, we keep all other parameters fixed at their baseline values.

Figure 7 compares the results for two models: the baseline heterogeneous-firm model and the representative-firm model. As the figure illustrates, state dependence is considerably stronger in the baseline model. Two reasons explain the difference. First, in the heterogeneous-firm model, what matters, besides average cash holdings, is the mass of firms on either flat part of the cash-holdings policy function—the constraint at the bottom or the satiation point at the top. In the representative-firm model, either constraint rarely binds and the MPPO remains below one. Second, cash fluctuations over the cycle are much more muted in the representative-firm model because the firm can more easily insure against aggregate fluctuations. The idiosyncratic productivity component creates an additional risk against which firms seek to insure and generates consistently greater dispersion and fluctuation of cash holdings over the cycle.

In Appendix E.1, we present further evidence from GIRFs following a negative TFP shock in the baseline and representative-firm models. The GIRFs indicate that state dependence is three times stronger in the heterogeneous-firm model.

Uncertainty shocks We introduce a Bloom-style aggregate uncertainty shock through a two-state process $S_t \in \{L, H\}$, using the quarterly transition matrix in Bloom et al. (2018). The high-uncertainty state widens the aggregate-TFP grid by a factor of 1.6, while leaving the transition matrix unchanged: when $S_t = L$, $A \in \{0.980, 1.020\}$, whereas when $S_t = H$, $A \in \{0.968, 1.032\}$. This is a pure second-moment shock. The level of aggregate TFP is unchanged on impact, but the dispersion of next period's TFP increases. The exercise tests whether the model's precautionary-cash mechanism also responds to changes in uncertainty, rather than only to first-moment productivity shocks.[28]

Appendix E.2 describes the implementation and results in more detail. The main finding is that uncertainty activates the same mechanism as first-moment shocks. Higher uncertainty increases firms' demand for liquidity, reduces the marginal propensity to pay out, and raises aggregate volatility. Thus, the model's central state-dependence result also applies to uncertainty-driven fluctuations.

5.4 Fiscal policy

We now use the model to study the aggregate effects of government transfers when firms hold different amounts of internal liquidity. The key object of interest is how the transmission of fiscal transfers depends on firms' payout behavior, summarized by the MPPO, and how this interaction generates nonlinear output responses over the business cycle.

Transfer process Government transfers follow a two-state Markov process that is independent of aggregate productivity. Let $T_t \in \{T_B, T_G\}$ denote the tax paid by the representative household at time t and equally distributed among firms. We calibrate the size

[28] The aggregate state space now becomes $X' = (\mathbf{G}(\Phi, A; N), A', S')$.

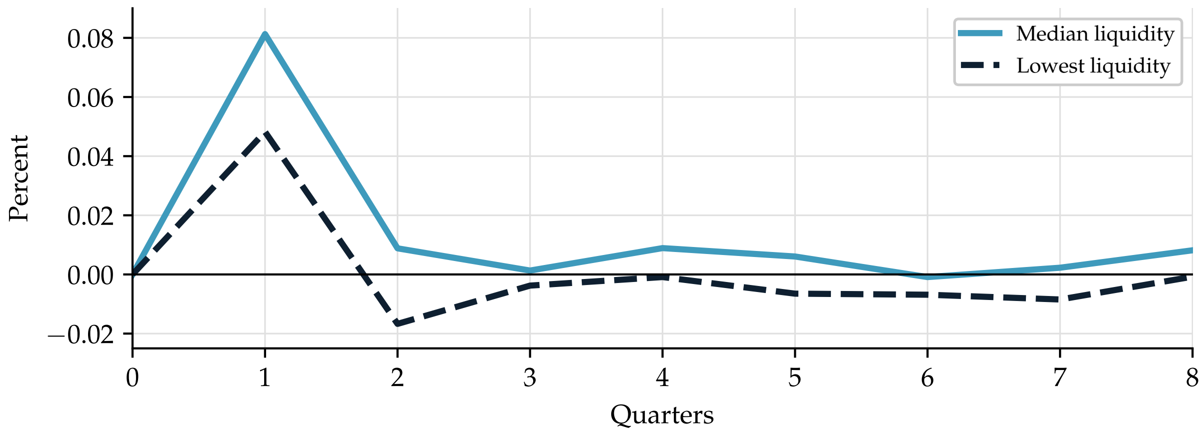

Figure 8: Output response to a transfer during a recession

Notes: The figure plots the generalized impulse response of output, in percent, to the same one-period transfer when aggregate TFP is low. The light-blue solid line starts from the median-liquidity distribution and the dark-blue dashed line from the lowest-liquidity distribution in the simulation.

of the fiscal intervention to match the scale of the *Troubled Asset Relief Program* (TARP). Specifically, we set the high-transfer state T_G to equal 2.4% of steady-state output, corresponding to the initial \$350 billion tranche authorized in late 2008 relative to aggregate GDP:

$$T_B = 0 \qquad \text{and} \qquad T_G = 2.4\% Y_{ss}.$$

The transition matrix governing the transfer process is given by

$$\Gamma_T = \begin{bmatrix} p(T_{t+1} = T_B \mid T_t = T_B) & 1 - p(T_{t+1} = T_B \mid T_t = T_B) \\ 1 - p(T_{t+1} = T_G \mid T_t = T_G) & p(T_{t+1} = T_G \mid T_t = T_G) \end{bmatrix},$$

with

$$p(T_{t+1} = T_B \mid T_t = T_B) = 0.98, \qquad p(T_{t+1} = T_G \mid T_t = T_G) = 0.$$

Hence, transfers arrive as rare and transitory fiscal expansions: once the economy enters the high-transfer state T_G, it deterministically reverts to the no-transfer state in the subsequent period.[29] This specification isolates the short-run propagation of fiscal policy and abstracts from long-lived wealth effects.

Subsidy effect Figure 8 plots the generalized impulse response function (GIRF) of aggregate output to a one-period transfer shock, conditional on the initial distribution of

[29]The aggregate state space now becomes $X' = (\mathbf{G}(\Phi, A; N), A', T')$.

firm cash holdings. We report responses starting from two states: a low-cash economy, in which a large fraction of firms operate close to their cash constraint, and a median-cash economy, in which firms hold substantially more internal liquidity.

The figure shows that output responds significantly more when aggregate cash holdings are higher. In contrast, when firms start from a low-cash state, the output response is muted and short-lived. The mechanism underlying this result follows from firms' endogenous payout behavior and the household labor-supply decision. When transfers are paid, they initially accrue to firms' balance sheets. Whether these resources affect real activity depends on how much of the transfer ultimately remains on the firm side of the economy.

When firms hold little cash, many of them operate at the cash constraint and exhibit a high MPPO. In this region of the policy function, firms behave like hand-to-mouth households: any additional liquidity that reaches the firm is immediately distributed as dividends. In equilibrium, the transfer therefore circulates from households to the government, from the government to firms, and back to households through higher net payouts, either as greater distributions or reduced equity issuance. Because resources immediately return to the household sector, household wealth is largely unaffected and labor supply changes little. As a result, the output response is small.

In contrast, when firms enter with high cash holdings, more firms are in the accumulation stage and the MPPO is low. Firms optimally retain a substantial fraction of the increased liquidity in the form of additional cash rather than distributing it immediately. In this case, fiscal transfers effectively reallocate resources away from households and into firms' balance sheets. Households become temporarily poorer and increase labor supply in order to smooth consumption. The resulting increase in labor supply leads to a rise in equilibrium output.

This mechanism also clarifies the limiting cases. If the MPPO were equal to one for all firms, transfers would be neutral: resources taken from households would be returned one-for-one through net payouts, leaving household wealth, labor supply, and output unchanged. Real effects arise precisely because the MPPO is strictly below one for a non-negligible mass of firms, allowing fiscal policy to affect the timing of resource use.

An important distinction between this mechanism and standard fiscal policy channels is that transfers do not involve wasteful spending or direct distortions. Resources are not destroyed nor inefficiently used; instead, they are temporarily held on firms' balance sheets. By preventing these resources from immediately flowing back to households and being used for consumption today, firms effectively delay their use over time. The output response therefore reflects an intertemporal reallocation driven by endogenous payout policies, rather than inefficient government expenditure.

The GIRF evidence highlights a central implication of the model: the effectiveness of fiscal policy depends critically on the endogenous distribution of firm liquidity. Transfers are most powerful when firms are liquid and payout behavior is muted, and least effective when firms are cash-poor and behave in a hand-to-mouth fashion. This state dependence is a direct consequence of the nonlinear cash policy function and would be absent in models with linear or exogenous payout behavior.

Taken together, these results show that corporate cash holdings are not merely a buffer against shocks but also a key determinant of fiscal transmission, shaping when and how government transfers translate into real economic activity.

6 Concluding remarks

This paper studies how the accumulation of corporate liquidity reshapes aggregate fluctuations. We develop a heterogeneous-firm business cycle model in which firms hold precautionary cash to hedge against convex equity issuance costs. Because the price of corporate cash has limited general equilibrium effects, firm-level nonlinearities survive aggregation and generate endogenous state dependence in macroeconomic dynamics.

We show that corporate liquidity fundamentally alters the transmission of both productivity and fiscal shocks. High cash holdings dampen consumption volatility by allowing firms to smooth payouts. Fiscal policy operates through the same channel: transfers are most effective when firms are liquid and retain resources, inducing a labor-supply response rather than immediate consumption.

All the results originate in a novel non-monotonicity in firms' liquidity policies. Costly external equity creates a precautionary motive that fades as buffers build, so firms stop accumulating at a finite target, while that target declines with productivity and internal cash flow rises. Their interaction generates three regimes: low-productivity firms that cannot build their desired buffers, intermediate-productivity firms that actively accumulate, and high-productivity firms that distribute marginal resources. The distribution across these regimes, summarized by the aggregate MPPO, transmits firm-level nonlinearities to aggregate consumption. Equilibrium prices do not smooth these nonlinearities, allowing state dependence to survive aggregation.

Empirical evidence supports each step of the model's mechanism. At the firm level, exploiting policy-induced liquidity variation from the 2002 and 2009 expansions of net operating loss carrybacks, we estimate an MPPO that declines sharply with profitability. The model closely reproduces this profile, together with untargeted moments on the distribution of corporate liquidity and the dividend-smoothing role of cash. At the household level, we show that corporate payouts pass through to consumption, connecting firms' liquidity decisions to aggregate demand. Corporate liquidity is therefore not merely a passive buffer against shocks but an active determinant of business-cycle dynamics and fiscal policy effectiveness. Understanding fluctuations requires accounting for firms' balance-sheet decisions, not only their investment choices.

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